# AVL Tree Rotations & Operations  Data Structures

> **Definition:** An **AVL Tree** (Adelson-Velsky and Landis) is a **self-balancing Binary Search Tree** in which the **Balance Factor (BF)** of every node is either **-1, 0, or +1**.

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## 1. Detailed Technical Explanation

### Balance Factor Formula:

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Balanc

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## 2. The Four AVL Tree Rotations

### 1. LL Rotation (Single Right Rotation)

- **Cause:** Insertion into the **Left subtree of the Left child** of node `z` (BF of `z` = +2, BF of left child = +1).

- **F

```

[ z ] (+

### 2. RR Rotation (Single Left Rotation)

- **Cause:** Insertion into the **Right subtree of the Right child** of node `z` (BF of `z` = -2, BF of right child = -1).

- **F

```

[ z ] (-2

### 3. LR Rotation (Double Rotation: Left then Right)

- **Cause:** Insertion into the **Right subtree of the Left child** of node `z` (BF of `z` = +2, BF of left child = -1).

- **F

```

[ z ] (+

#### 4. RL Rotation (Double Rotation: Right then Left)

- **Cause:** Insertion into the **Left subtree of the Right child** of node `z` (BF of `z` = -2, BF of right child = +1).

- **F

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### 3. Time Complexity of AVL Tree Operations

| Operation | Average Case | Worst Case |

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### 4. Must-Write Points for Exams

- The height of an AVL tree with `N` nodes is strictly bounded by `1.44 * log2(N)`.

- AV

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### 5. Quick Recall Flow

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Balanc
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